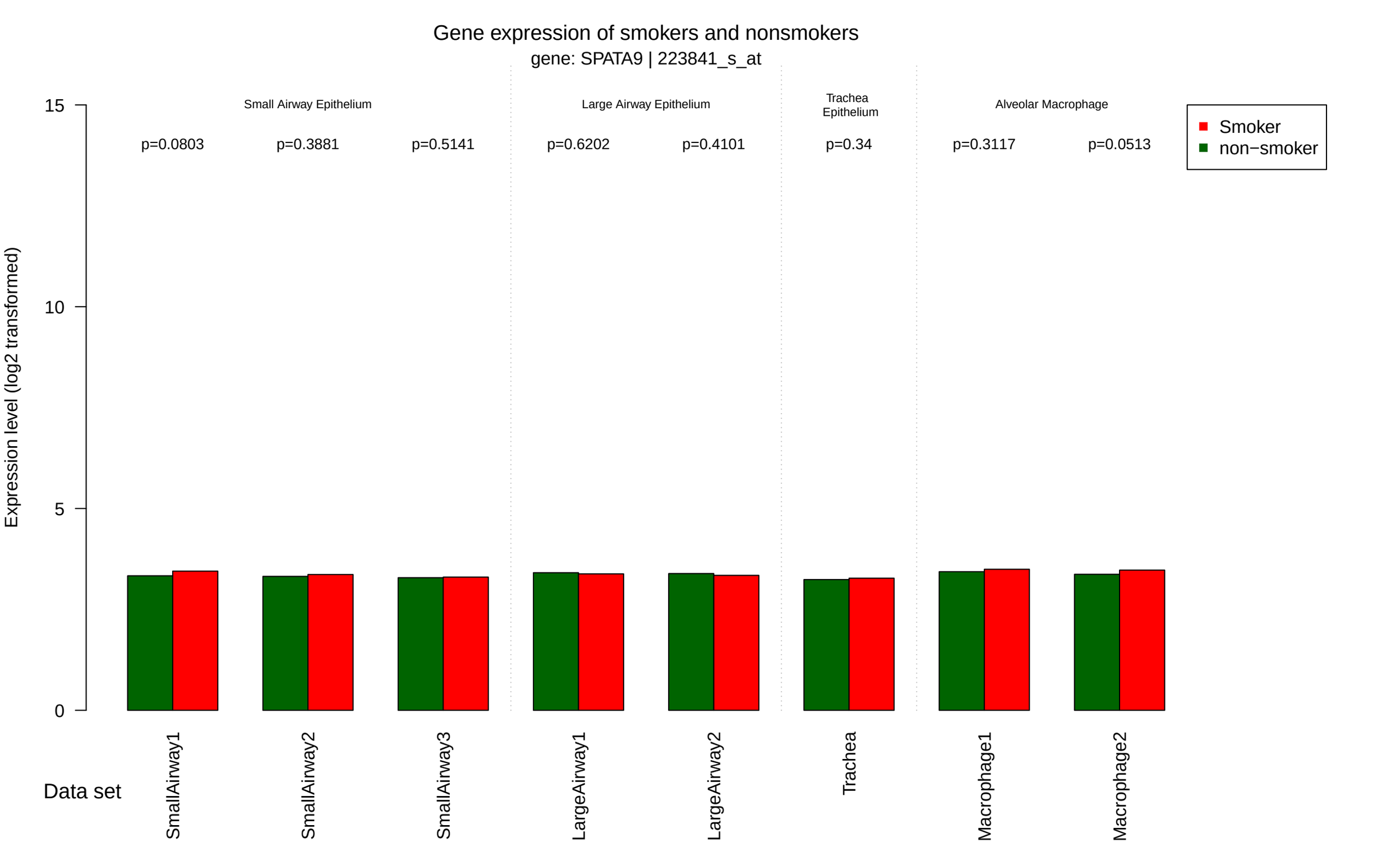
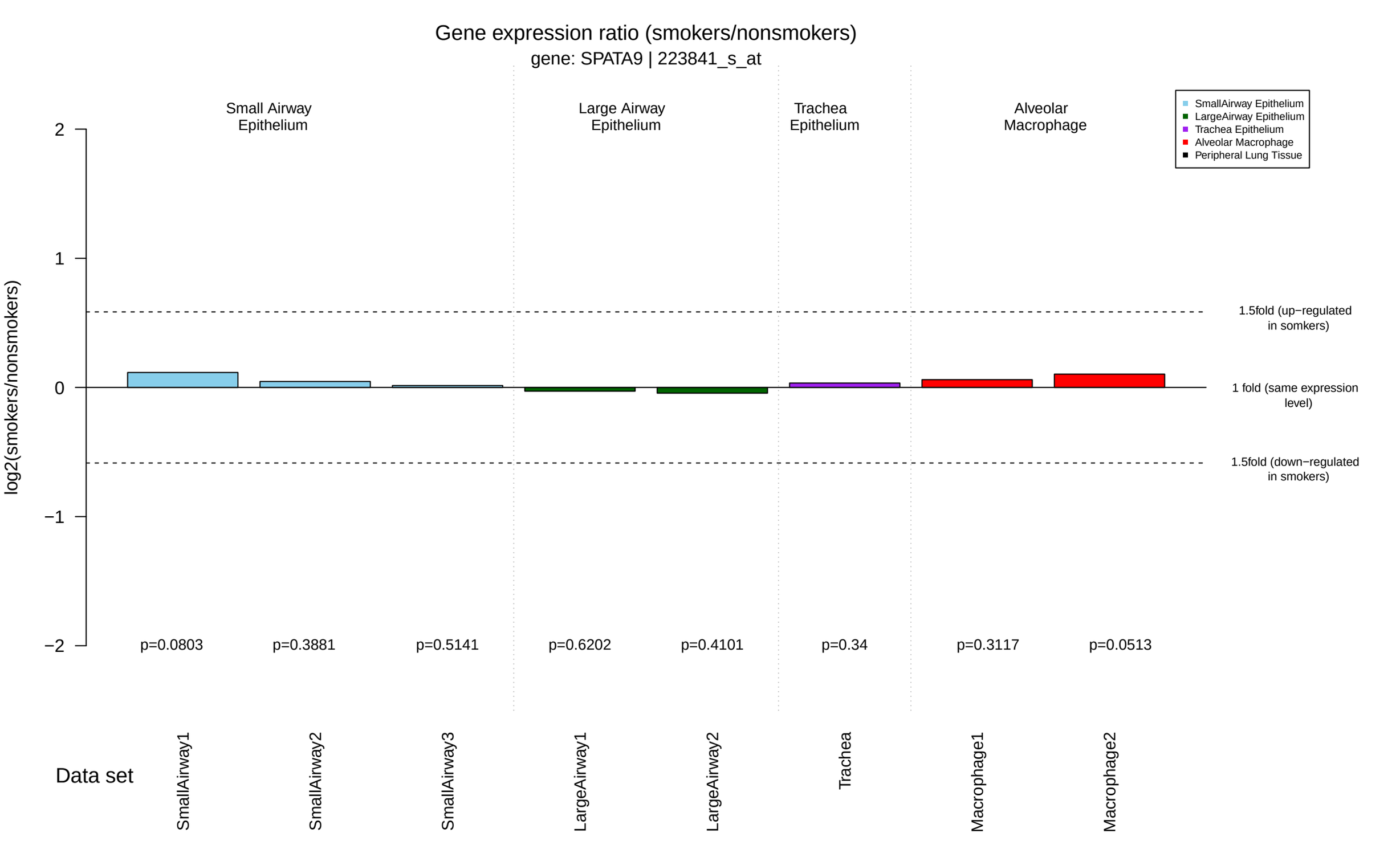
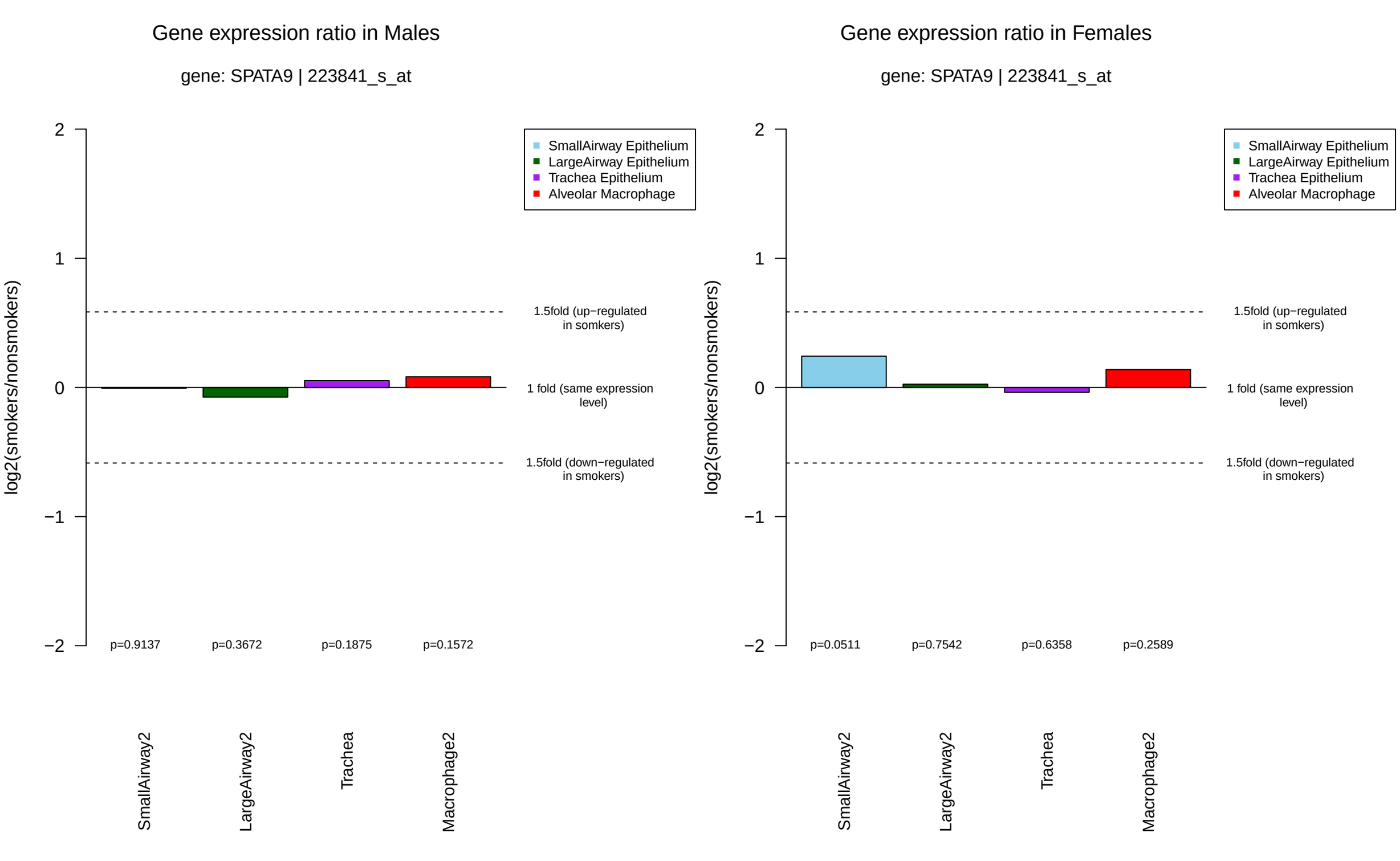
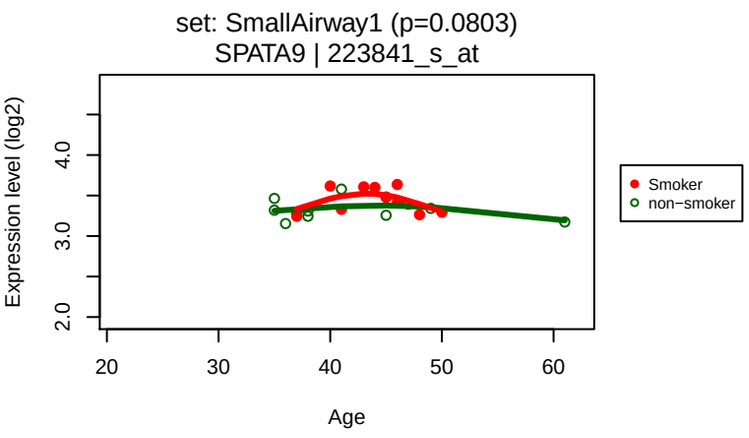
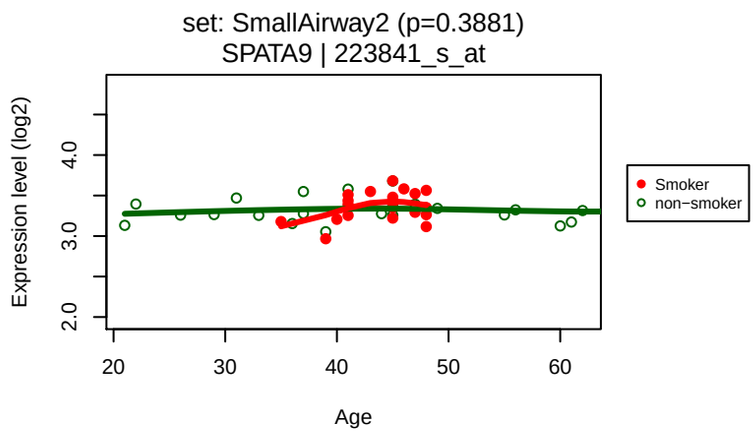
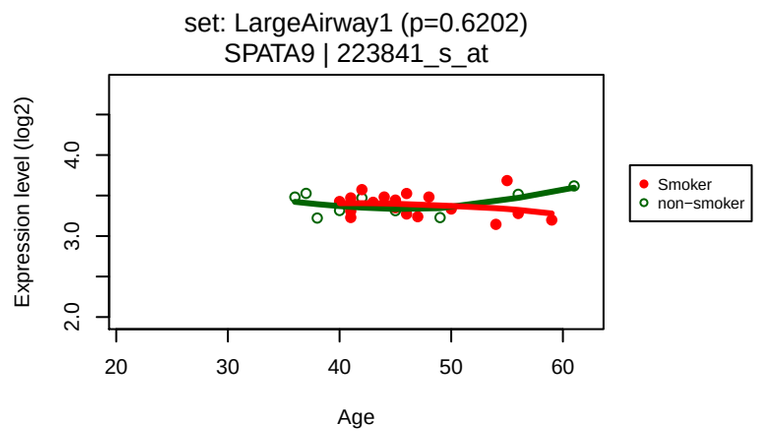

Gene expression of smokers and nonsmokers
gene: SPATA9 | 223841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0803
p=0.3881
p=0.5141
p=0.6202
p=0.4101
p=0.34
p=0.3117
p=0.0513
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

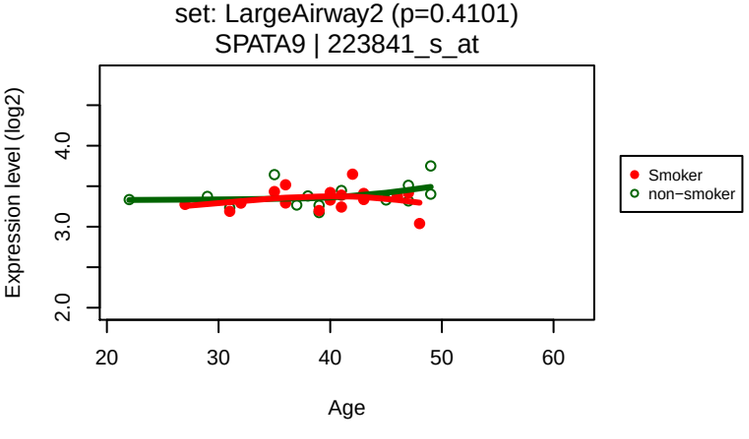

Gene expression ratio (smokers/nonsmokers)
gene: SPATA9 | 223841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0803
p=0.3881
p=0.5141
p=0.6202
p=0.4101
p=0.34
p=0.3117
p=0.0513
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

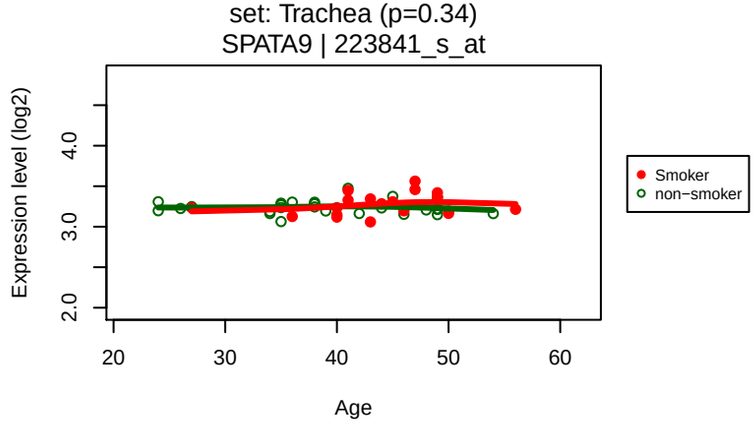

Gene expression ratio in Males
gene: SPATA9 | 223841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9137
p=0.3672
p=0.1875
p=0.1572
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA9 | 223841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0511
p=0.7542
p=0.6358
p=0.2589
SmallAirway2
LargeAirway2
Trachea
Macrophage2

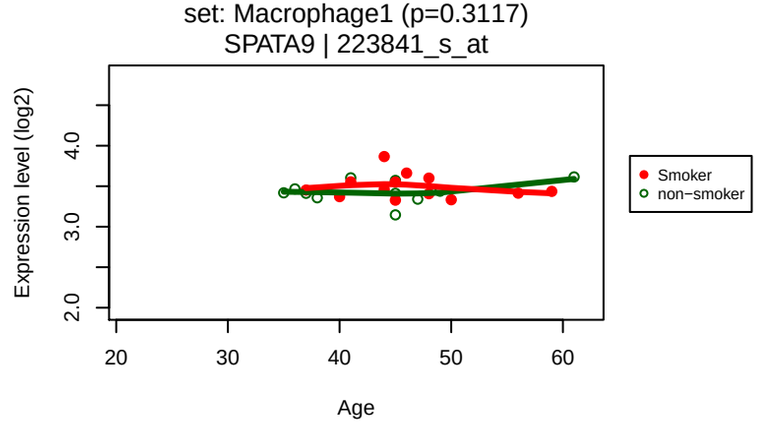

set: SmallAirway1 (p=0.0803)
SPATA9 | 223841_s_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

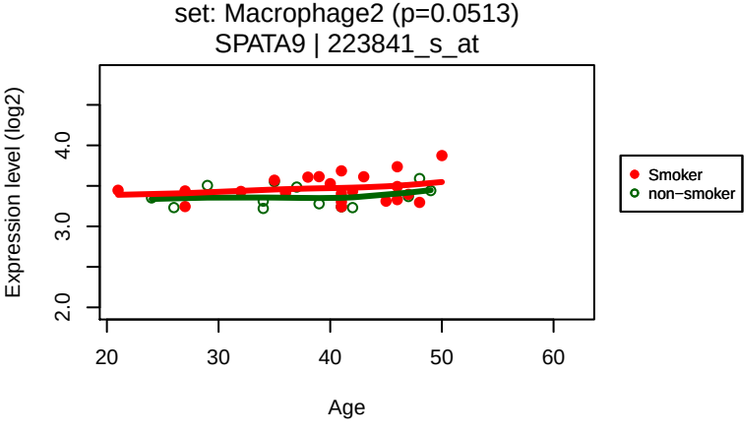

set: SmallAirway2 (p=0.3881)
SPATA9 | 223841_s_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

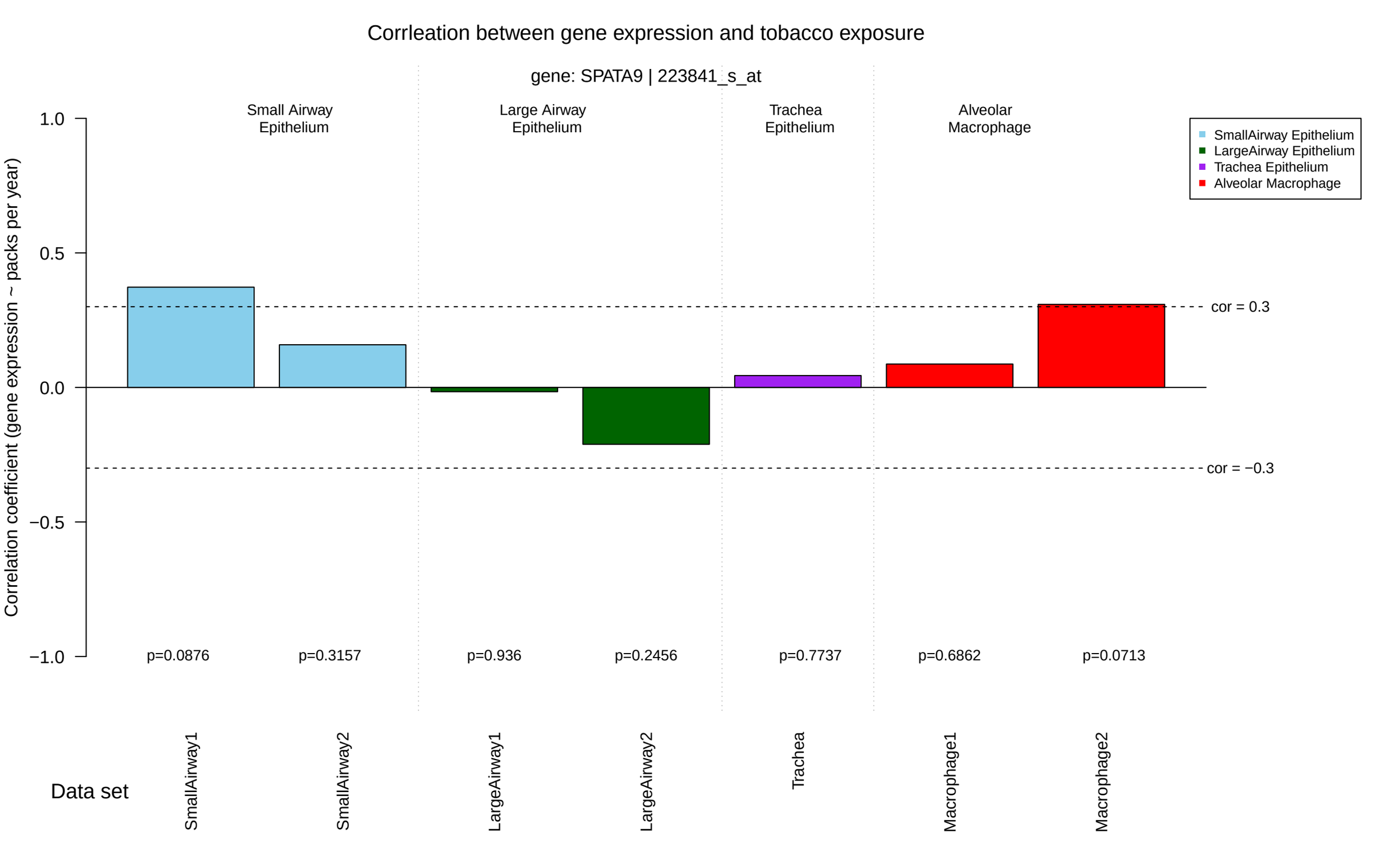

set: LargeAirway1 (p=0.6202)
SPATA9 | 223841_s_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4101)
SPATA9 | 223841_s_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.34)
SPATA9 | 223841_s_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3117)
SPATA9 | 223841_s_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0513)
SPATA9 | 223841_s_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA9 | 223841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0876
p=0.3157
p=0.936
p=0.2456
p=0.7737
p=0.6862
p=0.0713
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2